



Experiment No. 8

Aim – To implement Backtracking algorithm

Details – We are given n distinct positive numbers and find all combinations of these numbers whose sum are m using Backtracking algorithm.

Sum of subsets Problem – Suppose we are given n distinct positive numbers and we desire to find all combinations of these numbers whose sum are m. This is called the sums of subsets problem. We considering a backtracking solution using the fixed tuple size strategy. In this case the element x_i of the solution vector is either on or zero depending on whether the weights w_i is included or not. The children of any node in the following are easily generated. For a node at level i the left child corresponds to $x_i = 1$ and the right to $x_i = 0$. A simple choice for the bounding functions is $B_k(x_1, \dots, x_k)$ true iff

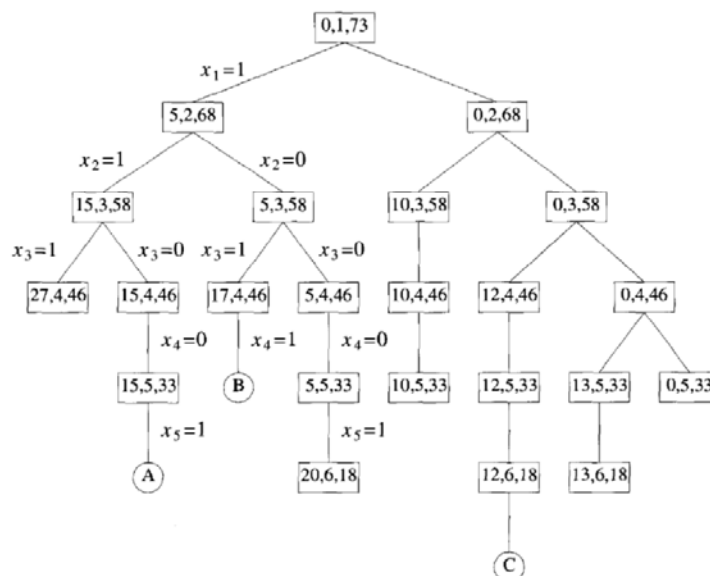
$$\sum_{i=1}^k w_i x_i + \sum_{i=k+1}^n w_i \geq m$$

$$\sum_{i=1}^k w_i x_i + w_{k+1} \leq m$$

Since our algorithms will not make use of B_n we need not be concerned by the appearance of w_{n+1} in this function. Although we have now specified all that is needed to directly use either of the backtracking schemas, a simpler algorithm results if we tailor either of these schemas to the problem at hand. This simplification results from the realization that if $x_k = 1$ then follows:

$$\sum_{i=1}^k w_i x_i + \sum_{i=k+1}^n w_i > m$$

For $W=\{5,10,12,13,15,18\}$ and $m=30$. Following recursion tree shows the working of Sum of subsets:



Important Links:

1) YouTube Video: [Sum of subsets](#) 2) Reading Resource: [Sum of subset](#) 3) Section 7.3 of Chapter 7 of Ref 2 Book

Input – Enter the weights of n number of elements and the integer m

Output – Print all possible subsets so that the sum is m.

Submission –

- 1) C/C++ source code of implementation
- 2) Verified output for the written source code with multiple inputs
- 3) One page report of Exp. 8